

COMPARISON OF SOME COMPUTER VISION BOARDS

	Arduino Nicla Vision	OpenMV Cam H7	HuskyLens	VC Picosmart	iVS-AWV3-AR0521
Processor	STM32H757AI16 Dual Arm Cortex M7/M4	ARM 32-bit Cortex-M7 CPU	Kendryte K21 (AI Chip with 64bit RISC-V dual core CPU)	FPGA module, High-end FPU processor	Allwinner V3 system on module
Camera/ image sensor	GC2145 (2MP colour camera)	OV7725 image sensor (can be interfaced with other sensors too)	HuskyLens – OV2640 (2MP camera), HuskyLens PRO – OV5640 (5MP camera)	1-Megapixel CMOS sensor	ON Semiconductor AR0521 sensor module
Notable characteristics and features	MP34DT06JTR digital MEMS microphone, LSM6DSOXTR 6-axis IMU, NXP SE050C2 Crypto chip	Double-precision floating point unit (FPU), Hardware JPEG encoding	It can be used as a stand-alone product or as a peripheral for a microcontroller or microcomputer	VC realtime-OS for real-time image processing	H.264, 1080p60 video codec, integrated ISP supports up to 8MP with auto image function
Connectivity and communication	Bluetooth, Wi-Fi, SPI, I2C, UART	Wi-Fi, BLE, SPI, I2C, UART, CAN	UART, I2C	FPC connector	Multiple wireless connectivity options available
Supported battery/ power source	Li-ion/LiPo single cell, 3.7V	3.7V LiPo battery	Two power pins on the 4-pin JST connector, micro USB connector	N/A	N/A
Suitable for	Asset tracking, object recognition, predictive maintenance	Object detection, line detection, video recording, QR code detection/decoding, eye tracking, colour tracking, frame differencing	Face recognition, object tracking, object recognition, line tracking, colour recognition, tag recognition	Object recognition, position control, barcode reading, Web edge and level control	Autonomous vehicle and aftermarket home automation camera systems
Compatibility/ integrations	Arduino IDE, Arduino CLI, Arduino IoT Cloud, OpenMV IDE	Edge impulse CLI, OpenMV IDE	Mind+ Editor, Arduino IDE	N/A	N/A
Primary programming language	C/C++	Python	C/C++	N/A	N/A
Edge capability	Yes	Yes	Yes	Yes	Yes
Industry-ready?	Yes	Yes	More suited to prototyping	Yes	Yes
Dimensions in mm (l x w x h)	22.86 x 22.86	45 x 36 x 30	52 x 44.5	22 x 23.5	N/A
Weight	2gm	19gm	N/A	N/A	N/A
MSRP	\$115.00	\$65.00	\$54.90	Quote based	Quote based
Datasheet	https://docs.arduino.cc/static/592480018567f53345d4f2b18199cd67/ABX00051-datasheet.pdf	https://cdn.shopify.com/s/files/1/0803/9211/files/OpenMV-H7.pdf?10726770933929369108	https://wiki.dfrobot.com/HUSKYLENS_V1.0_SKU_SEN0305_SEN0336	https://www.vision-components.com/en/products/oem/arm-linux/board-cameras/vc-picosmart/	https://www.iensoc.com/product/ivs-awv3-ar0521/

to embedded boards. Here are a few examples:

- CPUs
- GPUs
- Heterogenous systems (CPUs + GPUs + others)
- FPGAs
- ASICs
- Microcontroller boards
- Embedded systems (smart cameras)

So where do development boards stand?

Traditional computer vision systems used a camera connected to a general-purpose computer, which

used to run the application. In a smart camera though, the whole system can be integrated, right from capturing the image to processing to the final application. Today, the distinction between smart cameras and development boards has slowly started to fade, since many development boards have inbuilt cameras and are tailored for computer vision applications.

With more and more applications being deployed on edge, it makes more sense to deploy such devices on development boards. Here are some critical features that you need to consider while select-

ing a development board for your computer vision application:

- High processing speed
- Edge processing ability
- Faster access to memory
- Low power consumption
- Portable
- Inexpensive

While there are embedded platforms specifically for AI like NVIDIA Jetson and Raspberry Pi Compute, AI accelerators like Google TPU, SOMs/SBCs containing high-quality imaging sensors, we will focus on a few boards that were designed for computer vision and compare some of their features.